AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM270
Artificial intelligence

Final Examination
Spring Semester 2025/2026

Date: 2/June/ 2026

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 180 minutes

Instructions:

- Total Marks: **100**
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: This part consists of 10 multiple choice questions. For each question, encircle the label a, b, c, or d as correct answer. Each question carries 2 marks.

1 ..... creates large changes in genomes between generations; it is therefore an excellent method of introducing widespread diversity into the gene pool.

A Selection B Mutation
C mating pool D Crossover

2 In LSTM, the …… gate decide what we are going to output

A input B forget
C update D output

3 A..... is a way of identifying some part of a genome, a template.

A chromosome B genome
C schema D protein

4 If value of the learning rate is.... then learning takes place slowly: the weights move in small steps towards correct values.

A medium B small
C large D None of the above

5 ..... refers to changes made to an organism's genes at any point in an organism's life

A genes B Mutation
C crossover D protein

6 ..... is to use an object characteristics to identify which class/category it belongs to.

A training B Classification
C Clustering D None of the above

7 ..... is the distance from the first to the last fixed position in the schema.

A dimension B defining length
C Order D None of the above

8 in …... Architecture units are arranged in a single sheet, with excitatory and inhibitory connections among them. Units tend to excite other units in their neighborhood, and inhibit those further away.

A Recurrent B Lattice
C Layered D None of the above

9 In LSTM, the Ö... gate decide what new information we are going to store in the cell state.

A forget B input
C output D update

10 If value of the learning rate is ……. then learning takes place slowly: the weights move in small steps towards correct values.

A large B medium
C small D None of the above

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Part II: This part consists of four questions with a total of 80 Marks, solve all questions.

Question 2 [20 Marks]

A. Given the following solutions from GA population with their fitness:[10 marks]

A 23.000000
B 19.000000
C 30.000000
D 23.000000
E 29.000000

a. Calculate the probability of selecting each solution
b. Draw the roulette wheel representing the solution set
c. Assume the following stream of random numbers:[0.145859,0.718058,0.287476,0.603067,0.468134,], what will be the solutions entering the mating pool?

B. Having the following two chromosomes:[5 marks]

00100011101, 00011101011

Apply crossover at point 9.

C. Given genome A:0100000 [5 marks]

Write the output of the genome A after applying the mutation operator. Assume a mutation probability Pm= 0.29 and given the following sequence of random numbers corresponding to each chromosome:0.070020, 0.524147, 0.778167, 0.071494, 0.661345, 0.434867, 0.394183,

Question 3 [20 Marks]

Given the following particle positions in a PSO minimization optimization process at time t of the optimization:[15 marks]

X_1	X_2	X_3	X_4
0	-1	0	2
-1	2	-1	2
1	0	-2	1
2	-1	2	0

and given that each particle's attached velocity is:

V_1	V_2	V_3	V_4
-1.6	1.4	0.4	-1.0
-0.0	-0.7	0.1	-0.3
-0.9	1.3	-0.0	0.5
1.3	-1.3	-0.6	0.3

and given each particle's local best as:

P_1	P_2	P_3	P_4
1	-1	0	0
-1	2	-2	1
-2	-1	1	1
0	-1	1	-1

assuming the global best is:[1,-1,-2,0,]

Complete the following missing items that are calculated at time t+1 of the optimization.

Assume $\Phi_1=0.6$ and $\Phi_2=0.3$

The function to be minimized is:$3x_1^1+8x_2^4+6x_3^5+9x_4^6$

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V_1= -0.700000 0.000000 -3.600000 -0.500000 V_2= 2.000000 -1.600000 0.100000 -1.000000 V_3= 0.700000 -0.500000 1.800000 -1.800000 V_4=.....	X_1=..... X_2= 1.000000 0.400000 0.100000 -2.000000 X_3= 0.700000 -1.500000 -0.200000 0.200000 X_4= -0.500000 0.200000 0.600000 -0.300000
P_1= -0.700000 -1.000000 -2.600000 1.500000 P_2= -1.000000 2.000000 -1.000000 -1.000000 P_3=..... P_4= -0.500000 0.200000 0.600000 -0.300000	P_g=.....

b. Explain the concept of competitive learning and describe how it is used to train neural networks in vector quantization networks. [5 marks]

Question 4 [20 Marks]

A.

Given the weight matrix for layer 1 in a neural net as: 0.14 0.84 0.53 0.66 0.85 0.81 0.42 0.78 0.68 0.98 0.85 0.66 0.16 0.24 0.80 0.10 0.04 0.90 0.54 0.55 0.76 0.74 0.31 0.16 0.87 0.32 1.00 0.68 0.91 0.27 0.24 0.38 0.46 0.57 0.94	The weight matrix for output layer: 0.99 0.42 0.06 0.15 0.88 0.65 0.39 0.92 0.05 0.72 0.60 0.22 0.03 0.75 0.04 0.93 0.29 0.47 0.54 0.39 0.74 0.59 0.75 0.96 0.41 0.95 0.87 0.00
and given training points: 3 1 2 1 2	with its corresponding desired output: 0.3 0.9 0.9 0.1

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and assuming a learning rate of 0.7. Using backpropagation algorithm to update the neural weights for 1 iteration. Assume an activation function of type 1/(1+exp(-net)) for both layers

i. The net output on node 7 in the hidden layer = ...
ii. The actual output on node 7 in the hidden layer = ...
iii. The net output on node 4 in the output layer = ...
iv. The actual output on node 4 in the output layer = ...
v. ei on node 3 in the output layer = ...

Assume that the actual outputs on nodes 1,2,3,4,5,6 in the hidden layer are: 0.99,1.00,1.00,0.98,1.00,0.99 in order.
Assume that actual output on the output nodes 1,2,3 in the output layer are: 0.97,0.96,0.97 in order.

b.

Given the weight matrix for layer 1 in a neural net as: 0.56 0.46 0.10 0.32 0.88 0.02 0.94 0.73 0.08 0.62 0.59 0.46	The weight matrix for output layer: 0.19 0.34 0.40 0.50 0.08 0.55 0.99 0.99 0.33 0.97 0.46 0.54 0.23 0.23 0.98 0.13
and given training points: 1 2 1	with its corresponding desired output: 0.9 0.1 0.4 0.6

and assuming a learning rate of 0.9. Using backpropagation algorithm to update the neural weights for 1 iteration. Assume an activation function of type 1/(1+exp(-net)) for both layers

I. What is the weight change for weights attached to node 4 in the output layer= .....
II. What is the updated weights for weights attached to node 4 in the output layer= .....
III. What is ej on node 1 in the hidden layer= .....
IV. What is the weight change for node 3 in the hidden layer= .....
V. What is the updated weight for node 3 in the hidden layer layer= .....

Assume that the actual outputs on nodes 1,2,3,4 in the hidden layer are: 0.83,0.89,0.92,0.91 in order.
Assume that ei on the output nodes 1,2,3,4 in the output layer are: 0.02,-0.06,-0.05,-0.03 in order.
Assume that ej on the nodes 2,3,4 in the hidden layer are: -0.01,-0.01,-0.01 in order.

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Question 5 [20 Marks]

Given a neural network with first layer as RNN while the remaining layers are dense layers, assume the following:

For the RNN layer with linear activation:
W_{xh}=

0.53	0.81	0.12
0.50	0.80	0.62

W_{hh}=

0.44	0.14
0.26	0.27

For the dense layers:
W_2 with linear activation=

0.77	0.47
0.28	0.37
0.27	0.48

Given the following input data point X=

0.00	0.00
0.00	1.00
0.00	1.00

While the desired output y_d=

0.26
0.79
0.73

At a learning rate η =0.57, calculate the following:

a. Calculate each aggregate output and actual output for the RNN layer.
b. Calculate the aggregate and actual output for the dense layers.
c. Calculate e for each dense layer.
d. Calculate e at each step for the RNN layer.
e. Calculate ΔW for each dense layer.
f. Calculate ΔW_{xh}, ΔW_{hh} for each step in the RNN layer.
g. Calculate the aggregate ΔW for the RNN layer.

End of questions

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